



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.07
TUTOR GUIDE

The cases and materials in this guide were written and developed by;

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The cases and materials in this guide were reviewed by;

- € Members of the University of Notre Dame Clinical Years Curriculum Committee, critical care working group; Last Reviewed Dec 2010.

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RELEVANT PRIOR LEARNING

MED1000

- 1.01 First Aid Principles (L)
- 1.01 First Aid (CCS)
- 1.02 Principles of Fluid Replacement (L)
- 1.22 The critically ill patient (L)

MED2000

- 2.01 Basic Life Support (CCS)
- 2.01 Assessment & Management of sudden collapse (L)
- 2.02 Assessment & management burns patient (CCS)
- 2.08 Sudden Cardiac Death & AED (L)
- 2.08 Defibrillation in fatal arrhythmias (CCS)
- 2.08 Basic Life support algorithm (CCS)
- 2.09 Advanced Life Support algor (CCS)
- Rural trauma week (W)

MED 3000

- 3.02 Coronary artery disease (Short case tutorial & L)

WEEK 07 EMERGENCY MEDICINE		
Yr	Wk	LEARNING OBJECTIVES
4	7	- Know the key steps in the assessment and management of the non-responsive patient. - Revise the basic life support algorithm and be able to implement in an emergency situation. - Know the key elements of the advanced life support algorithm including defibrillation and administration of drugs. - Know the potentially reversible causes of cardiac arrest and the initial steps in management. - Summarise the current evidence for defibrillation in cardiac arrest. - Be able to develop a management plan for the immediate management of a patient following successful resuscitation post cardiac arrest. - Know the indications and the implementation of cooling post cardiac arrest. - Summarise the current evidence for the benefits of patient cooling post cardiac arrest.
4	7	- Know the key steps in the emergency assessment and management of a patient with suspected acute coronary syndrome. - Outline the current evidence for the use of reperfusion therapy and adjunct therapy associated with reperfusion in the management of patients with acute STEMI. - Outline the factors that influence the choice of reperfusion therapy and adjunct therapy associated with reperfusion in the management of patients with acute STEMI. - Outline the important serious complications of acute STEMI and explain the underlying pathophysiology. - Discuss the ethical issues that can arise when commencing and ceasing CPR on a patient. - Summarise the current evidence regarding improved outcomes in hospital cardiac arrests as a result of the implementation of hospital arrest [MET] teams.
4	7	- Revise the key ECG features of supraventricular tachyarrhythmia and in particular be able to distinguish between atrial fibrillation [AF] and supraventricular tachycardia [SVT]. - Outline the possible causes of AF and explain the underlying pathophysiology for each cause. - Outline the possible causes of SVT and explain the underlying pathophysiology for each cause. - Know the key steps in the acute assessment and management of a patient with SVT and AF, including indications and contraindications for treatments used, and be able to implement in an emergency situation.

WEEK 07 EMERGENCY MEDICINE		
Yr	Wk	LEARNING OBJECTIVES
4	7	• Know the key steps in preparations for the arrival of a patient with potentially serious / life threatening burns. • Know the key principles in the assessment and management of burns in the adult and paediatric patient. • Know the key components of the clinical assessment of burns in particular the assessment of extent and depth of the burn injury. • Know the key criteria for the transfer of a burn injury patient to a specialised burns unit. • Outline the principles of the short and longer term management of a burn injury patient.

CASE ONE

Short case number: 4_07_01

Category: Clinical skills

Discipline: Emergency Medicine

Setting: General Medical Ward_rural

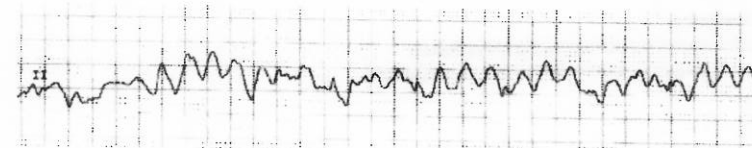
Topic: Cardiac Arrest

CASE

You are the Intern on evening overtime at Wagga Wagga Base Hospital. You receive a "Rapid Response" page from Ward 3, a General Medical Ward. On arrival at the patient bedside a few minutes later you discover the patient is a 66 year old male, who is lying in bed, pale/grey in colour and unresponsive. There are three nursing staff present.

You quickly check for responsiveness and signs of life. The patient does not open his eyes to verbal prompts, the chest wall is not moving, and there is no palpable carotid pulse.

The nursing staff have attached the cardiac monitor on the resuscitation trolley. The following rhythm is visible on the monitor screen.



QUESTIONS

1. Outline your immediate management plan for this patient?
2. Outline the key elements of the Basic Life Support Algorithm
3. Outline the key elements of the Advanced life support Algorithm, including Defibrillation, and drug Administration.
4. What are the potentially reversible causes of cardiac arrest and how would you assess and manage these?
5. What is the most important intervention in terms of favourable outcome of cardiac arrest? What is the evidence basis for this?
6. How would you acquire more information about this patient in order to determine possible causes of and potential outcomes of cardiac arrest?

FURTHER CASE INFORMATION

Following your interventions the patient returns to Sinus Rhythm, with a HR of 98, BP of 106/56, and is breathing spontaneously at a rate of 22 per minute. His GCS is 6/15 (E1V1M4).

His medical record reveals he has been admitted with a Non ST elevation Myocardial Infarction.

QUESTIONS

7. What are the main considerations in your post Cardiac Arrest management plan?
8. Discuss the evidence base, indications and procedure for Post Cardiac Arrest cooling.

Resources:

- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

Electronic Resources available on Blackboard:

- Holzer et al (2010). Hypothermia for Neuroprotection in adults after cardiopulmonary resuscitations (review). The Cochrane Collaboration. Retrieved from: <http://onlinelibrary.wiley.com/doi/cochrane/clsystrev/articles/CD004128/pdf fs.html>

Other Resources:

Basic Life Support Guidelines: www.resus.org.au

Advanced Life Support Guidelines: www.resus.org.au

ANSWERS

1. Outline your immediate management plan for this patient?

Commence Basic Life Support

- CPR

Call for Help

Ask Nursing staff to put out a "Cardiac Arrest" page and page the Medical Registrar / Senior staff member in the hospital

Request the Cardiac Arrest Trolley be brought to the patients room

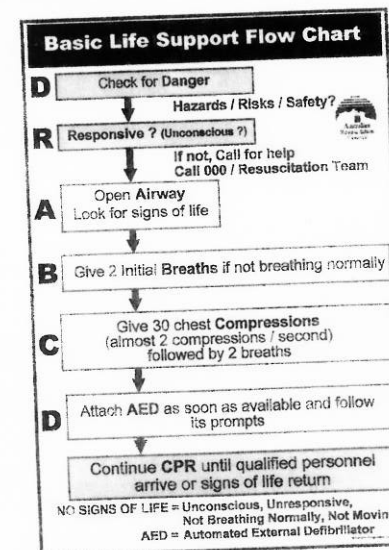
Assemble the arrest team and assign roles

- You currently have 4 staff members. Suggestions could include:
 - o Airway management
 - o CPR (2 staff alternating)
 - o Staff member to draw up and deliver medications

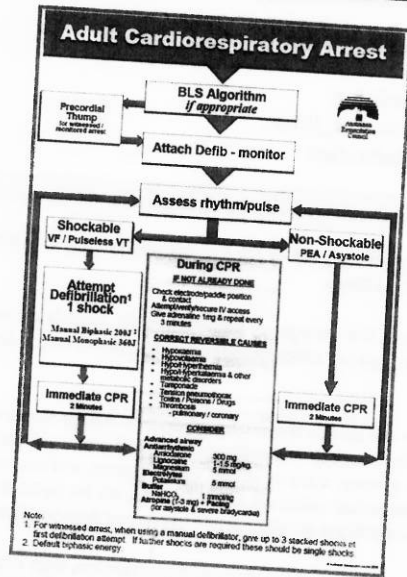
Commence Cardiopulmonary Resuscitation as per the advanced life support algorithm

- EARLY Defibrillation
- CPR
- Adrenaline 1mg IV every 3 minutes
- Consideration of Reversible Causes

2. Outline the Key elements of the Basic Life Support Algorithm



3. Outline the Key elements of the Advanced life support Algorithm, including Defibrillation, Drug Administration and Potentially Reversible Causes of Cardiac Arrest.



4. What is the most important intervention in terms of favourable outcome of cardiac arrest?
What is the evidence basis for this?

2005 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Science with treatment recommendations. Part 3: Defibrillation. *Circulation*, 2005; 112: III-17 – III-24.

- Early defibrillation for Ventricular Fibrillation is the most important intervention in terms of return of spontaneous circulation and survival to hospital discharge
- A period of 1 to 3min of CPR prior to Defibrillation in terms of improved return of spontaneous circulation, may be beneficial if there is a delay of >4 minutes from arrest to arrival of Medical Personnel

Hazinski et al. Lay rescuer automated external defibrillator ("Public Access Defibrillation") programs. Lessons learned from an international multicenter trial. *Circulation*, 2005; 111: 3336-3340

- Latest evidence include PAD trial which involved 993 facilities in North America, with 239 episodes of out of hospital cardiac arrest with attempted resuscitation (July 2000 – September 2003)
- Bystander only CPR, 15/107 patients survived to hospital discharge
- Bystander CPR plus Automated External Defibrillation 30/128 survived to discharge
- Statistically significant

- Nursing Staff – reason for admission, recent observations, current therapy
- Admission notes – admission documentation, medications charts
- Medical Record – past medical history, past admissions
- Patients Admitting Consultant – details of the current admission, progress
- Patients relatives / next of Kin – normal level of functioning, recent history

6. What are the main considerations in your post Cardiac Arrest management plan?

- Stabilisation of the patient – ABCDE
- Appropriate disposition of the patient – Need to transfer patient to a treatment area appropriate for his condition, i.e. Coronary Care Unit, Intensive Care Unit
- Identification of Cause and/or precipitants for Cardiac Arrest - Assessment and investigations for the cause, include history and progress, complete physical examination, investigations (bloods, CXR, Echocardiography)
- Management / Intervention – dependent on identified cause, i.e. PCA for STEMI

7. Discuss the evidence base, indications and procedure for Post Cardiac Arrest cooling.

Nolan et al (2003). Therapeutic hypothermia after cardiac arrest: an advisory statement by the advanced life support task force of the international committee on resuscitation. *Circulation*, 2003; 108: 118-121

2 prospective RCT published in 2002

Comparison of mild hypothermia with normothermia in comatose survivors of out of hospital arrest

Criteria for entry into study:

- Return of spontaneous circulation
- Patients remaining intubated and ventilated
- Persistent coma after out of hospital arrest due to VF
- Exclusion of patients with likely non-cardiac cause of arrest

European Study

- 55% patients in hypothermia group had favourable neurological outcome (75/136)
- 34% patient in normothermia group had favourable neurological outcome (54/137)
- 41% Deceased by 6 months in the hypothermia group (56/137)
- 55% Deceased by 6 months in the normothermia group (76/138)

Australian Study

- 49% patients in hypothermia group had good neurological function at discharge (21/43)
- 26% patients in normothermia group had good neurological function at discharge (9/34)
- 51% mortality at discharge in hypothermia group (22/43)
- 68% mortality at discharge in normothermia group (23/34)

"Favourable Neurological Outcome" – able to live independently and work at least part-time

Protocol

- Active cooling to between 32 – 34 degrees for 12 to 24 hours post Cardiac Arrest
 - o Cooling blankets
 - o Ice packs
 - o Fans
 - o Cooling helmet
 - o Infusion of Cold Normal saline (4 degrees)
 - o Reduce shivering – will lead to warming and increased oxygen consumption, use neuromuscular blockade and sedation (therefore must be intubated and ventilated)

Recommendations for Cooling

- Unconscious adult patients with spontaneous circulation after out-of-hospital cardiac arrest should be cooled to 32 to 34 degrees for 12 to 24 when the initial rhythm was VF
- Such cooling may also be beneficial for other rhythms or in-hospital cardiac arrest

CASE TWO

Short case number: 4_07_02

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department_Urban

Topic: Acute Myocardial Infarction

CASE

You are the Emergency Resident assigned to the Resuscitation bay at St Vincent's Hospital Emergency Department. It is 1100hrs.

A 58 year old male presents to the Emergency Department via Ambulance with Chest pain and Shortness of Breath. The Ambulance officer hands over the patient to the Emergency Registrar who is also present.

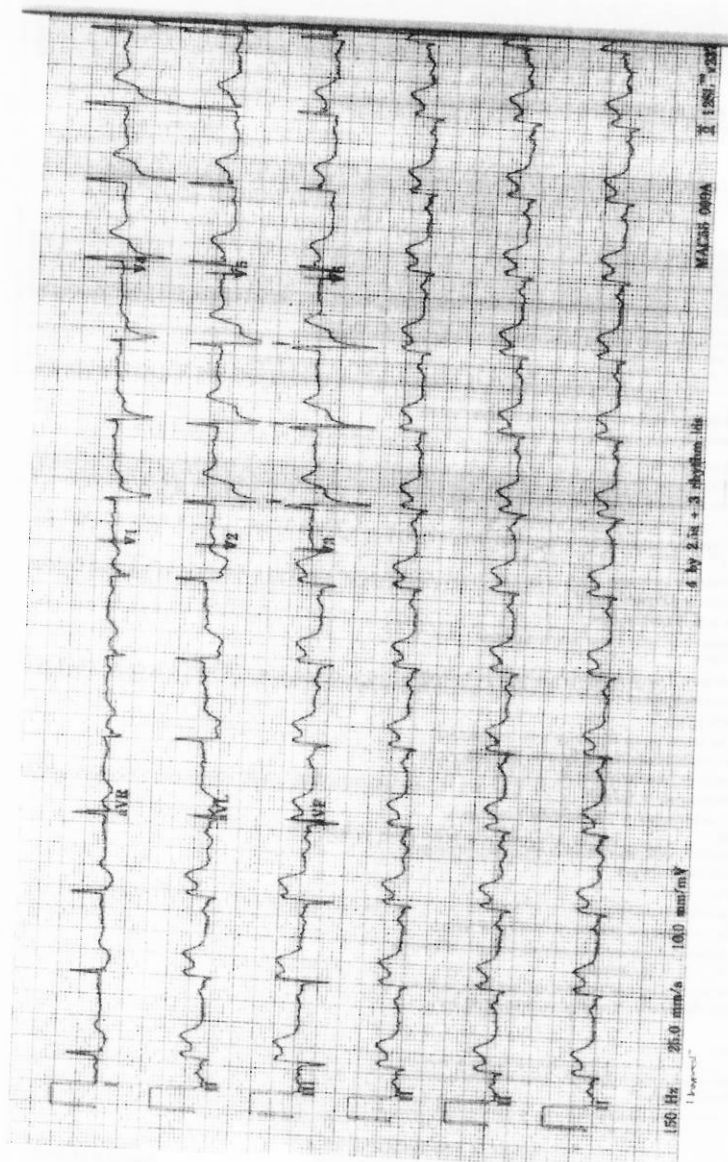
The patient was working in the garden this morning when he had an acute onset of chest pain. The pain was 10/10, crushing, central, and radiated to his left arm, neck and jaw. When the ambulance arrived he was sweaty, dizzy on sitting up, nauseous, and has vomited once. Observations on ambulance arrival were: HR 110, BP 90/45, RR 30, SaO₂ 92% on Room Air, Chest Pain 8/10. Ambulance officers administered oxygen and Aspirin 300mg PO.

Ambulance officers tell you he has a past history of Hypertension, High cholesterol, and smoking. He has not had chest pain before. His current medications include Ramipril 10mg Mane, Aspirin 100mg Daily, and Atorvastatin 40mg Nocte. He has no allergies.

On arrival to Emergency your assessment is as follows:

- A Patent and protected
- B RR 32, SaO₂ 98% on 15L Oxygen via a Non re-Breather Mask
Increased Work of breathing evident with accessory muscle use. Nil cyanosis
Equal chest wall rise. Decreased air entry bilaterally, widespread crepitations throughout both lung fields.
- C HR 120 sinus rhythm. BP 94/50. CR = 3 sec. Diaphoretic. Pale. Systolic murmur heard on Auscultation. ECG in progress
- D GCS 14, PEARL, BSL 6.0
- E Temperature 35.8

You quickly place 2x intravenous cannula, send bloods for EUC, FBC, LFT, CMP, Troponin, CK, and order an urgent mobile CXR. The nurse hands you the ECG.



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SCHOOL OF MEDICINE SYDNEY TUTOR

11-Feb-11

QUESTIONS

1. Describe the ECG changes. What is the diagnosis based on the ECG?
2. Outline the possible causes for the patho-physiological abnormalities?
3. What are the options for the immediate management of this patient? Include discussion of the indications and contraindications of management plans.

FURTHER CASE INFORMATION

Following successful intervention, the patient is commenced on Inotropic support and is transferred to the Coronary Care Unit (CCU). You decide to visit the CCU to enquire as to how the patient is 2 days later.

While you are waiting to speak to the Cardiology Registrar the Arrest alarm sounds. You realise it is your patient, and rush to see if you can assist the team.

Following 30 minutes of Cardiopulmonary resuscitation and Advanced Life Support, the patient is declared deceased.

QUESTIONS

4. Outline the important serious complications post Acute Myocardial Infarction.
5. In the case of prolonged advanced life support and Cardiac Arrest, what factors would help you determine when to cease resuscitation?
6. What is the evidence that hospital arrest (MET) teams improve outcomes in hospital Cardiac arrest situations?

Resources:

- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

Electronic Resources available on Blackboard:

- Aroney et al (2006). Guidelines for the Management of Acute Coronary Syndromes 2006. MJA, 184(8), 17 Apr 2006.
- Part 2: Ethical Issues. Circulation. 2005; 112: IV-6-IV-11.

ANSWERS

Describe the ECG changes. What is the diagnosis based on the ECG?

Acute infero-lateral ST elevation AMI

Outline the possible causes for the physiological abnormalities?

Acute left ventricular failure and Cardiogenic Shock

What are the options for the immediate management of this patient? Include discussion of the indications and contraindications of management plans.

ABCDE

- Oxygen
- IV access
- Bloods sent
- Aspirin 300mg
- GTN sublingual (provided systolic BP >100)
- Analgesia – Morphine

REPERFUSION – call cardiology to organise urgent Angio +/- stenting
If angiography not available on site consider Thrombolysis

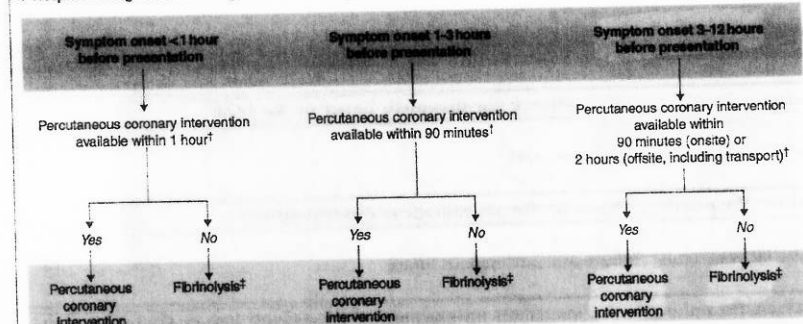
3 Summary of adjuvant therapy associated with reperfusion

Medication	Primary percutaneous coronary intervention	Fibrin-specific fibrinolytic
Aspirin	Yes	Yes
Clopidogrel	Yes (unless the need for acute CABG is likely)	Yes (unless the need for acute CABG is likely)
Heparin	Unfractionated heparin (ACT 200–300 s if using glycoprotein IIb/IIIa inhibitors, 300–350 s if not) ^a	Unfractionated heparin (APTT 1.5–2 times control [approx 50–70 s]) ^a or Enoxaparin ^a
Glycoprotein IIb/IIIa inhibitors	Abciximab optional	No

CABG – coronary artery bypass graft; ACT – activated clotting time; APTT – activated partial thromboplastin time.

^a Care should be taken in patients aged over 75 years, or those who have significant renal dysfunction – dose adjustment is required.

4 Hospital management of ST-segment-elevation myocardial infarction*



* Assuming no contraindications to fibrinolytic therapy – see Box 5. † Time delay refers to time from first medical contact to balloon. ‡ Patients with ongoing symptoms or instability should be transferred for PPCI.
Note: Reperfusion after 12 hours is indicated for cardiogenic shock, ongoing pain or haemodynamic instability (see text).

5 Contraindications and cautions for fibrinolysis use in ST-segment-elevation myocardial infarction*

Absolute contraindications

Risk of bleeding

- Active bleeding or bleeding diathesis (excluding menses)
- Significant closed head or facial trauma within 3 months
- Suspected aortic dissection (including new neurological symptoms)⁵⁰

Risk of intracranial haemorrhage

- Any prior intracranial haemorrhage
- Ischaemic stroke within 3 months
- Known structural cerebral vascular lesion (eg, arteriovenous malformation)
- Known malignant intracranial neoplasm (primary or metastatic)

Relative contraindications

Risk of bleeding

- Current use of anticoagulants: the higher the international normalised ratio (INR), the higher the risk of bleeding
- Non-compressible vascular punctures
- Recent major surgery (< 3 weeks)
- Traumatic or prolonged (> 10 minutes) cardiopulmonary resuscitation
- Recent (within 4 weeks) internal bleeding (eg, gastrointestinal or urinary tract haemorrhage)
- Active peptic ulcer

Risk of intracranial haemorrhage

- History of chronic, severe, poorly controlled hypertension
- Severe uncontrolled hypertension on presentation (> 180 mmHg systolic or > 110 mmHg diastolic)
- Ischaemic stroke more than 3 months ago, dementia, or known intracranial abnormality not covered in contraindications

Other

- Pregnancy

* Modified with permission of the American College of Cardiology and the American Heart Association.¹²

Outline the important serious complications post Acute Myocardial Infarction.

1. Arrhythmias and Conduction disturbances – VF, VT
2. Pericarditis – acute, Dresslers syndrome
3. Thromboembolism – formation in hypokinetic myocardium due to relative stasis
4. Acute mitral insufficiency – due to papillary muscle rupture
5. Ventricular septal defect
6. Cardiac Rupture

When should you consider withholding CPR? What factors are important in the decision to cease CPR?

Reference: Part 2: Ethical Issues. Circulation. 2005; 112:IV-6 – IV-11

Reasons to withhold CPR

- Valid Do Not Attempt Resuscitation (DNAR) order or advanced directive
- Signs of irreversible death – rigour mortis, decomposition, dependent lividity, decapitations
- Futility – No physiological benefit can be expected because vital functions have deteriorated despite maximal therapy

Reasons to Terminate CPR

Ultimately decision rests with the Treating physician in the hospital
Based on consideration of:

- Time to CPR
- Time to defibrillation
- Co-morbid disease
- Pre-arrest state
- Initial arrest rhythm – VF better prognostic outcome than asystole
- >30min downtime without any ROSC
 - o Any ROSC during resuscitation resets the clock to zero

Evidence that prolonged resuscitation >30 minutes shows extremely limited favourable outcomes

Exceptions/Controversies:

Severe Pre-arrest Hypothermia
Toxicological, i.e. Beta Blocker overdose
Electrocution Injury
Massive PE treated with thrombolysis

Reference: Van Walraven et al (2001). Clinical Decision aid to discontinue in-hospital CPR. JAMA, 2001; 285(12): 1602-1606.

All in hospital arrests between 1987-1996 - 1884 patients after exclusions
Decision Aid: Patient has chance of hospital discharge if ANY of the following is true

- i. Arrest was witnessed
- ii. Initial cardiac rhythm was VT or VF
- iii. Pulse was regained during the first 10 minutes of CPR

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Of the 327 resuscitations for which patients were discharged from hospital, all but 3 satisfied the decision aid (Sensitivity 99.1%). Of these three patients, one died at one month post discharge, second and third discharged to nursing home (need assistance with ADL's)

What is the evidence that hospital arrest teams improve outcomes in hospital Cardiac arrest situations?

MERIT Study Investigators. Introduction of the medical emergency team (MET) system: a cluster randomised controlled trial. Lancet, 2005; 366(9492): 1164.

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SCHOOL OF MEDICINE SYDNEY TUTOR

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CASE THREE

Short case number: 4_07_03

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department_Urban

Topic: Supra-ventricular Arrhythmias

CASE

You are the Intern working at Royal Melbourne Hospital Emergency Department. Two patients have presented via ambulance in the last 5 minutes. Both have been given a triage category 2 and placed in the Resuscitation bays.

PATIENT 1

A 68 year old male presents to the Emergency Department with Palpitations, chest pain and shortness of breath.

On arrival at triage his heart rate is 158 beats per minute. He is placed in the resuscitation bay, and nursing staff connect him to the monitors and obtain an urgent ECG. You undertake a rapid ABCDE assessment. You place an 18G IVC in the cubital fossa.

He states he has had similar episodes of these palpitations in the past.

HR 153, BP 105/66, CR = 2.5 sec, RR 24, SaO2 98% 15L Hudson Mask Oxygen

ECG 1:



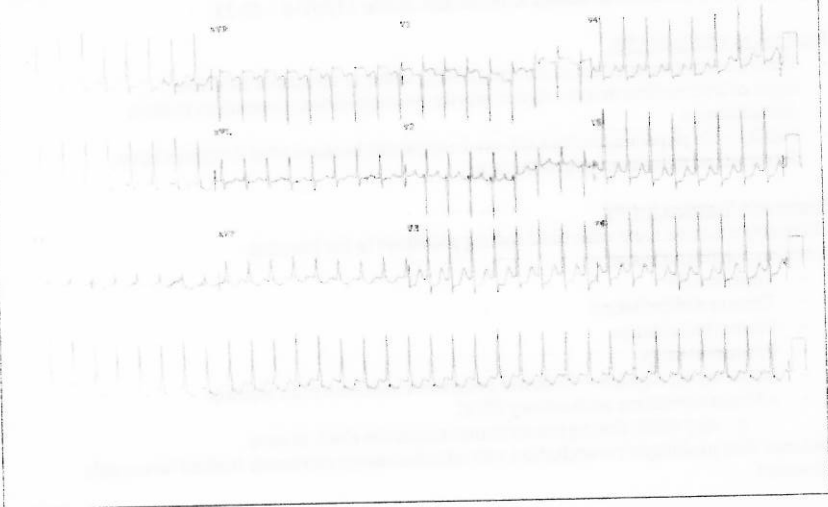
PATIENT 2

A 22 year old female presents with Palpitations and Shortness of breath. She states she has had multiple episodes of similar presentations in the past, the last being 6 months ago.

On arrival at triage her heart rate is 200 beats per minute. She is placed in the resuscitation bay and is connected to the cardiac monitor. The nursing staff are attending to Patient 1, and you are required to obtain an urgent ECG. You also undertake a rapid ABCDE assessment, administer oxygen, and place an 18G cannula in the cubital fossa.

HR 180, BP 115/72, RR 20, SaO2 99% on 6L Oxygen

ECG 2:



QUESTIONS

PATIENT 1

1. Describe the changes on the ECG. What is the diagnosis?
2. What are the criteria for distinguishing between types of supra-ventricular tachycardias?
3. Outline the causes/pathophysiology of this type of arrhythmia.
4. Outline when immediate intervention would be required in this condition? Does this patient require immediate intervention?
5. Outline the options for management for this condition
6. Outline your management plan for this patient.

PATIENT 2

1. Describe the changes on the ECG. What is the diagnosis?
2. Outline the possible causes of this type of arrhythmia.
3. Outline the options for the management of this patient, including the indications and contraindications of each management strategy.

Resources:

- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS**PATIENT 1**

Describe the changes on the ECG. What is the diagnosis?

Rapid Atrial Fibrillation

- irregular rhythm
- absence of P waves

What are the criteria for distinguishing between types of supra-ventricular tachycardias?

Rate

Rhythm – regular (sinus), irregular

QRS complex – narrow

Outline the causes/pathophysiology of this type of arrhythmia.

Table 5.4.3: Causes of AF seen in the Emergency Department

Cardiac

- Ischaemic heart disease
- Pericarditis
- Hypertension
- Rheumatic Heart Disease
- Pre-excitation syndromes
- Cardiomyopathy
- Atrial septal defect
- Atrial myxoma
- Postoperative

Non-cardiac

- electrolyte imbalance
- sepsis
- pulmonary embolism
- drug and alcohol intoxication
- chronic obstructive airways disease
- thyrotoxicosis
- lung cancer

Outline when immediate intervention would be required in this condition? Does this patient require immediate intervention?

DC Cardioversion would be required if the patient were haemodynamically unstable – hypotension, decreased LOC

Patient is conscious and has reasonable BP so cardioversion not immediately indicated

Outline the options for management for this condition

Management will be dependent on:

- Whether AF is intermittent or chronic
- Ventricular response rate / haemodynamic stability
- Underlying structural heart disease
- Associate medical conditions
- Age

Intermittant AF

- Choice is between DC cardioversion in ED (chemical or electrical) if time of onset of AF is certain (within 48 hours), or rate control with anticoagulation with delayed elective cardioversion, or rate control alone

Chronic AF

- aim is to rate control + treat any concurrent illness

Pharmacological Agents for Cardioversion

- No structural heart disease
 - o Flecainide (2mg/kg over 30min)
 - o Sotalol (1mg/kg over 30min)
- Structural Heart Disease present
 - o Amiodarone (2-3mg/kg over 30min – do not give if long QT)

Pharmacological Agents for Rate Control

- Beta blockers, i.e. Metoprolol
- Calcium channel blockers, i.e. Verapamil
- Amiodarone

Outline your management plan for this patient.

Need further history

- Ischaemic heart disease / structural heart disease
- Current medications
- Previous Cardiac Echocardiogram study results
- Previous episodes of AF management

Likely given age and previous episodes will require rate control + anticoagulation with view to perform ECHO to rule out intra-cardiac thrombus, then elective cardioversion.

PATIENT 2

Describe the changes on the ECG. What is the diagnosis?

Supraventricular Tachycardia

Outline the possible causes of this type of arrhythmia.

Ectopic Atrial focus
Re-entry circuit

Outline the options for the management of this patient, including the indications and contraindications of each management strategy.

Haemodynamically unstable

- requires urgent cardioversion

Haemodynamically stable

- attempt vagal manoeuvres
- if unsuccessful use pharmacological management
- adenosine 6-12mg rapid bolus IV OR verapamil 5mg IV slowly (need to ensure no structural abnormalities if using Verapamil)
- if resistant to chemical cardioversion may require electrical cardioversion

CASE FOUR

Short case number: 4_07_04

Category: Clinical Skills

Discipline: Emergency Medicine

Setting: Emergency Department_Urban Tertiary

Topic: Burns

CASE

You are the medical student during your emergency rotation. The department receives a BAT CALL. A 63 year old female presents to the emergency department via ambulance with burns. She was cooking at home when she slipped on the kitchen floor, knocking a pot of boiling water on the stove over and onto herself.

She arrives in the department. You see her as she is being transferred onto the bed.



The Emergency Medicine Registrar starts the ABCDE assessment and management sequence.

You consider the important aspects of the Assessment and Management of the patient who has sustained a burn injury.

QUESTIONS

1. Outline the preparation necessary prior to this patient arriving in the Emergency Department?
2. What aspect of the history and initial examination of the burns concerns you most in terms of ABCDE?
3. Outline the important principles of assessment and management of burns patients, including adult and paediatric patients.
4. Describe how to assess the depth and extent of burn injury.
5. Outline the criteria for transfer of this patient to a specialist burns unit
6. Describe the important principles of short to long term management of burns patients.

Resources:

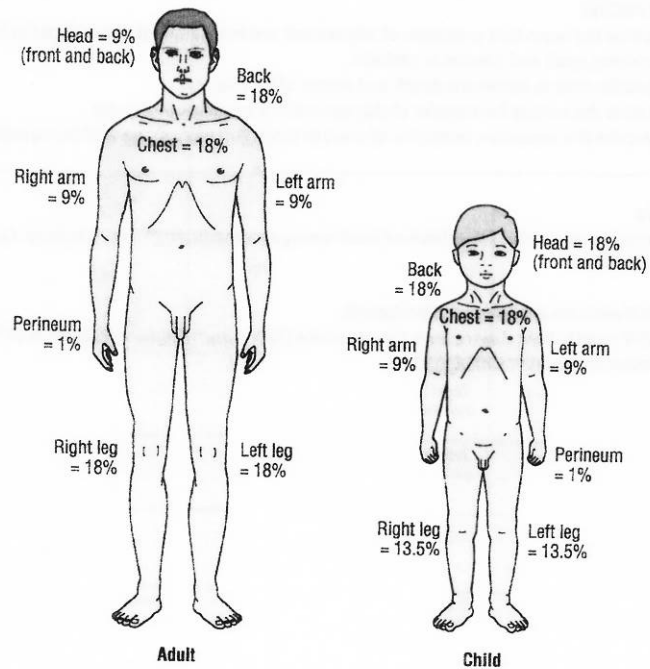
- Cameron et al (2009). Textbook of adult emergency medicine (3rd ed.). Sydney: Churchill Livingstone.

Electronic Resources available on Blackboard:

- NSW Health, NSW Severe Burn Injury Service (2008). Burn Transfer Guidelines (2nd Ed). Document number GL2008_012.

Outline how you would assess the depth and extent of burn injury.

Estimation of Total Burn Surface Area (TBSA): Rule of Nines



For smaller burns use the palmar surface of the Patients hand, which approximates 1% body surface area.

Depth

- Superficial
- Epidermis
- Heal spontaneously

Partial Thickness

- Destruction of epidermis, and superficial dermis.
- Blister formation
- Further classified into superficial and deep partial thickness
- Healing dependent on amount of intact epithelium

Electrical burns
Burns associated with other significant injuries
Circumferential burns of limbs or chest
Burns in any patients with pre-existing medical disorders which could complicate management, prolong recovery or affect mortality

Describe the important principles of immediate to short to long term management of burns patients.

IMMEDIATE TO SHORT TERM

FIRST AID

Stop burning process / remove source of burning
Cool burn
ABCDEF
Analgesia
Wash burn – sterile saline
Dress burn – cling wrap, simple dressing, bactigras + light bandage
Elevation burn area if possible – i.e. limb, head and neck
Special Areas – catheterisation early in genital burns
Escharotomy – for circumferential burns with circulatory compromise

SHORT TO MEDIUM TERM

Intubation and ventilation for facial/neck/airway burns – until swelling subsides
Analgesia
Burns dressings – Anti-bacterial, i.e. Acticoat
Debridement, excision and skin grafting
Assessment and early treatment infection
Nutrition

MEDIUM TO LONG TERM

Skin grafting
Infection control
Physiotherapy
Occupational Therapy
Rehabilitation

1. The first part of the paper is devoted to a general discussion of the problem of the existence of a solution of the system of equations (1) for arbitrary values of the parameters α and β .

2. In the second part, the problem of the existence of a solution of the system of equations (1) for arbitrary values of the parameters α and β is solved. It is shown that the system of equations (1) has a solution for arbitrary values of the parameters α and β if and only if the condition

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